	the ions when melting.
3	
(a)(i)1.	m is the mass of a single gas molecule.
(a)(i)2.	$\langle c^2 \rangle$ is the mean squared speed of the gas molecules.
(a)(i)3.	$\frac{1}{2}m\langle c^2 \rangle$ is the mean kinetic energy of a gas molecule.
(a)(ii)	$U = \frac{3}{2}N_A kT$ $= \frac{3}{2}(6.02\times10^{23})(1.38\times10^{-23})(273.15+27.0)$ OR $U = \frac{3}{2}nRT$ $= \frac{3}{2}(1)(8.31)(273.15+27.0)$
	$= 3740$ J
(b)	At the same temperature, average kinetic energy of the molecules are the same. $\frac{1}{2}m_N c_{rms,N}^2 = \frac{1}{2}m_O c_{rms,O}^2$ $\frac{c_{rms,N}}{c_{rms,O}} = \sqrt{\frac{m_O}{m_N}}$ $= \sqrt{\frac{5.31\times10^{-26}}{4.65\times10^{-26}}}$
	$= 1.07$
(c)(i)	Work done by the gas = $p\,\Delta V$ $= (8.00\times10^6)(7.10-3.00)\times10^{-5}$
	$= 328$ J
(c)(ii)	$Q = mc\Delta\theta$ $= (1.27\times10^{-3})(1.01\times10^3)(900)$
	$= 1.15443\times10^3$ J $= 1.15\times10^3$ J
(c)(iii)	$\Delta U = W + Q$ $= -328 + 1154$ $= 826$ J
4	